Programming Fundamentals

Lab 3 Submission	
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Roll No	| TO BE ADDED |
Program	| BSCS – 1st Semester |
Section	(E)
Submitted To	|Sir Asad Majeed|
Date	|15 – 10 – 26 |

Lab Task 01

Assume that the following variables are defined:

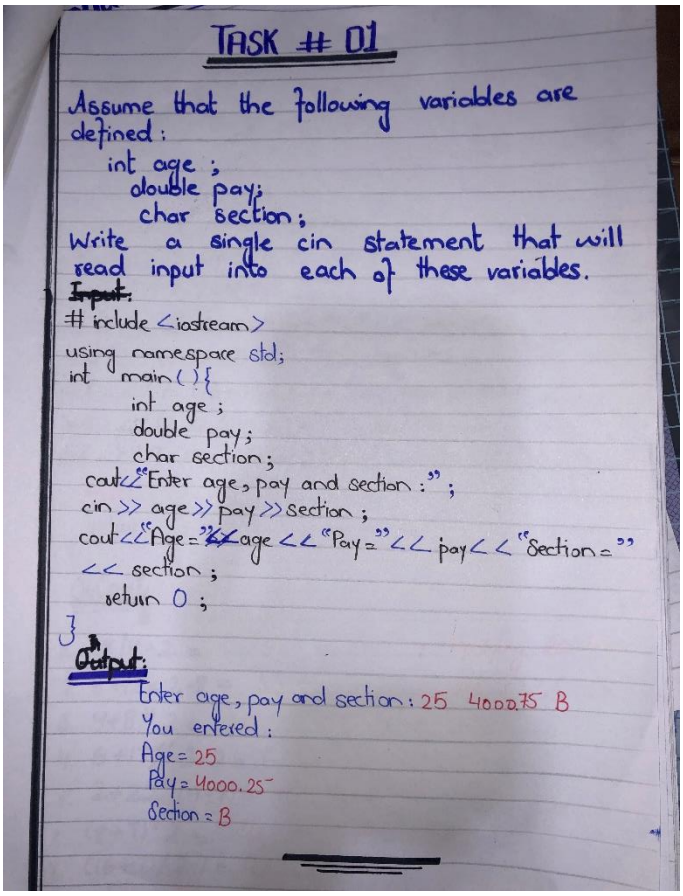
int age;

double pay;

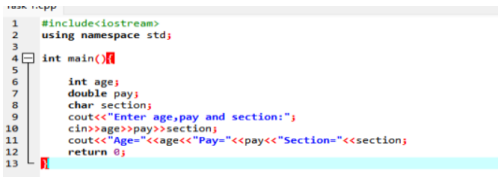
char section;

Write a single cin statement that will read input into each of these variables.

HANDWRITTEN :



CODE On Desktop :



```
#include<iostream>
using namespace std;

int main(){

    int age;
    double pay;
    char section;
    cout<<"Enter age,pay and section:";
    cin>>age>>pay>>section;
    cout<<"Age="<<age<<"Pay="<<pay<<"Section="<<section;
    return 0;
}
```

OUTPUT :

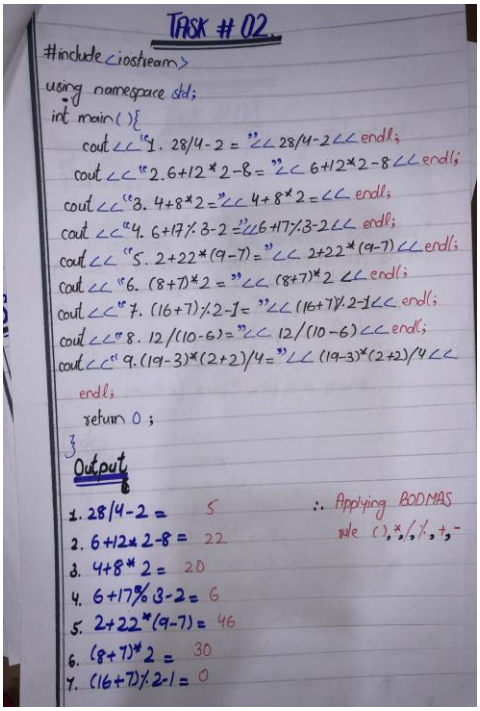
```
Enter age,pay and section:25 4000.75 B
Age=25Pay=4000.75Section=B
--------------------------------
Process exited after 56.92 seconds with return value 0
Press any key to continue . . .
```

Lab Task 02

Complete the following table by writing the value of each expression in the Value column

according C++ language rules.

HANDWRITTEN :



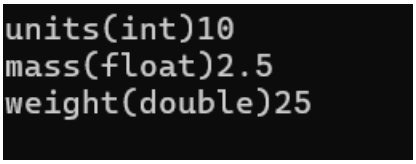
CODE On Desktop :

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << 28 / 4 - 2 << endl;
    cout << 6 + 12 * 2 - 8 << endl;
    cout << 4 + 8 * 2 << endl;
    cout << 6 + 17 % 3 - 2 << endl;
    cout << 2 + 22 * (9 - 7) << endl;
    cout << (8 + 7) * 2 << endl;
    cout << (16 + 7) % 2 - 1 << endl;
    cout << 12 / (10 - 6) << endl;
    cout << (19 - 3) * (2 + 2) / 4 << endl;

    return 0;
}
```

OUTPUT :



LAB TASK 03

Assume a program has the following variable definitions:

int units;

float mass;

double weight;

weight = mass units;

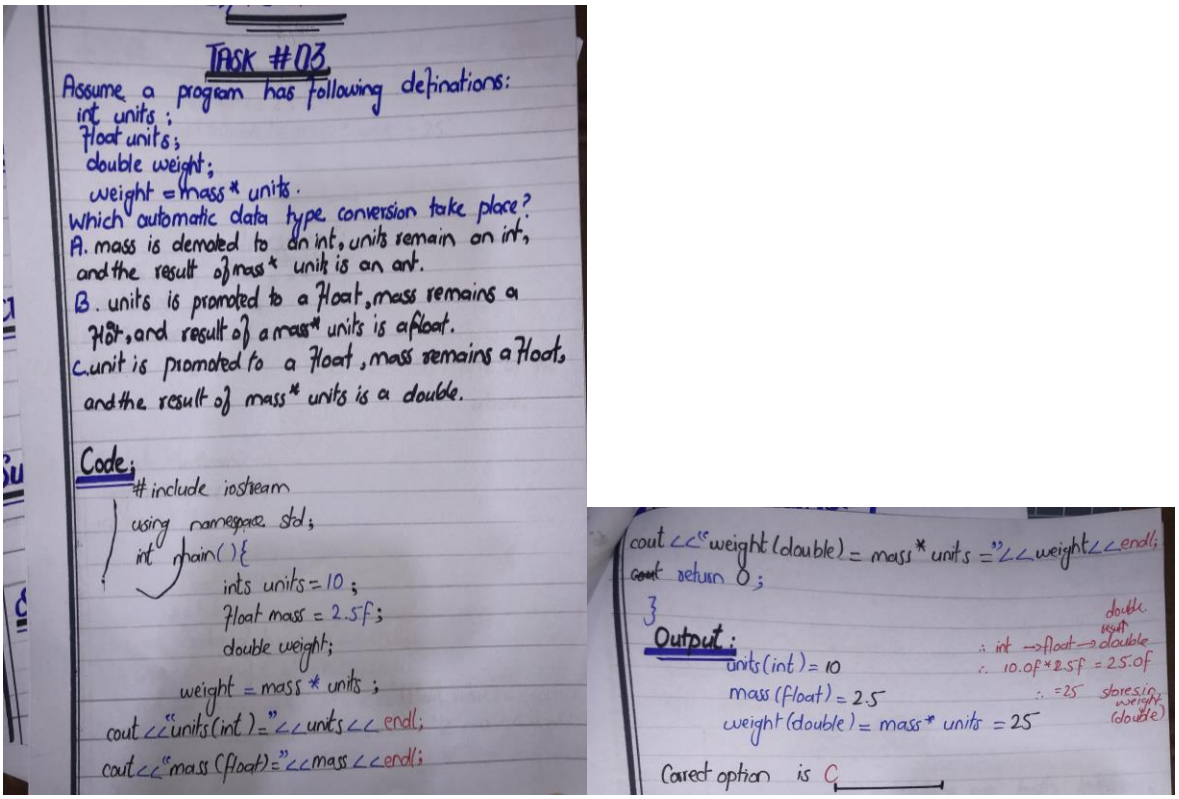
Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.

B. units is promoted to a float, mass remains a float, and the result of mass • units is a float.

C. units is promoted to a float, mass remains a float, and the result of mass * units is double

HANDWRITTEN :



CODE On Desktop :

Task 4.cpp

```cpp
#include<iostream>
using namespace std;
int main(){
    int units=10;
    float mass=2.5f;
    double weight;
    weight=mass*units;
    cout<<"units(int)"<<units<<endl;
    cout<<"mass(float)"<<mass<<endl;
    cout<<"weight(double)"<<weight<<endl;
    return 0;
    }
```

OUTPUT :

```
units(int)10
mass(float)2.5
weight(double)25
```

LAB TASK 04

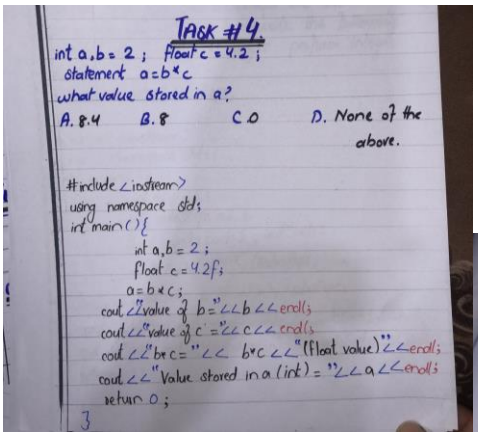
Assume a program has the following variable definitions:

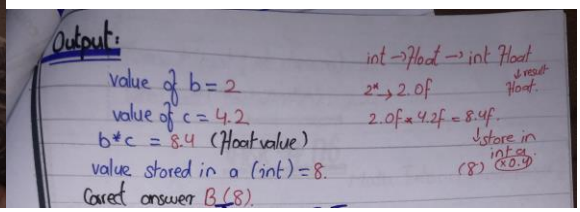
int a, b = 2; float c = 4.2;

and the following statement: a = b * c;

What value will be stored in a?

HANDWRITTEN :





CODE On Desktop :

task 4.cpp

```cpp
#include<iostream>
using namespace std;
int main(){
    int a,b=2;
    float c=4.2f;
    a=b*c;
    cout<<"value of b="<<b<<endl;
    cout<<"value of c"<<c<<endl;
    cout<<"b*c="<<b*c<<"(float value)"<<endl;
    cout<<"value stored in a(int)"<<a<<endl;
    return 0;
}
```

OUTPUT :

```
value of b=2
value of c4.2
b*c=8.4(float value)
value stored in a(int)8
```

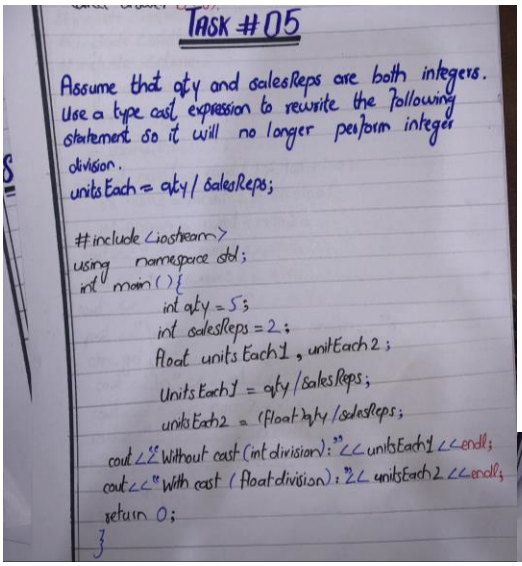
LAB TASK 05

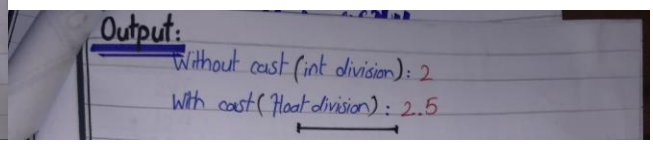
Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the

following Statement so it will no longer perform integer division.

unitsEach qty / salesReps;

HANDWRITTEN :





CODE On Desktop :

```cpp
#include<iostream>
using namespace std;
int main(){
    int qty=5;
    int SalesReps=2;
    float unitsEach1,unitsEach2;
    unitsEach1=qty/SalesReps;
    unitsEach2= (float)qty/SalesReps;
    cout<<"Without cast(int division):"<<unitsEach1<<endl;
    cout<<"with cast(float division)"<<unitsEach2<<endl;
    return 0;
}
```

OUTPUT :

```
Without cast(int division):2
with cast(float division)2.5

--------------------------------
```

LAB TASK 06

Math Tutor: (hint rand())

Write a program that can be used as a math tutor for a young student. The program should

display two random numbers to be added, such as

247

+ 129

The program should then pause while the student works on the problem. When the student

is ready to check the ansvver, he or she can press a key and the program will display the

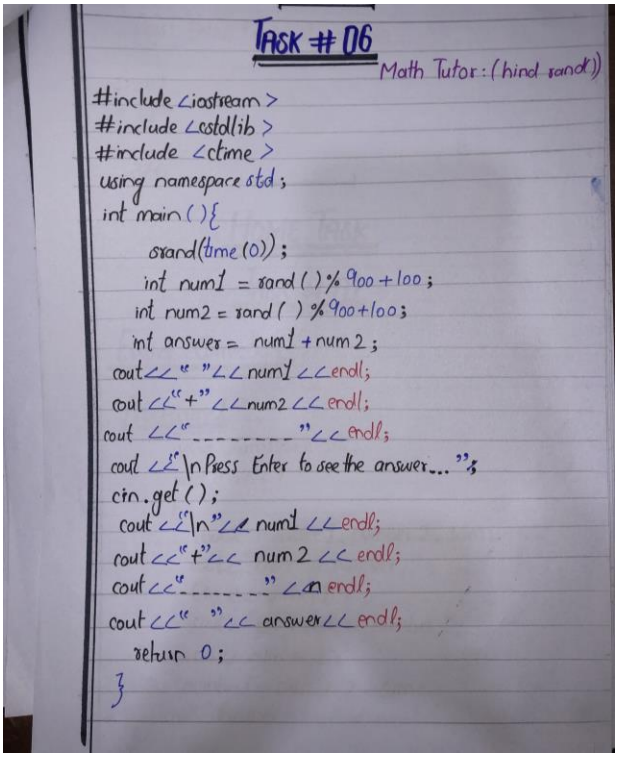
correct solution:

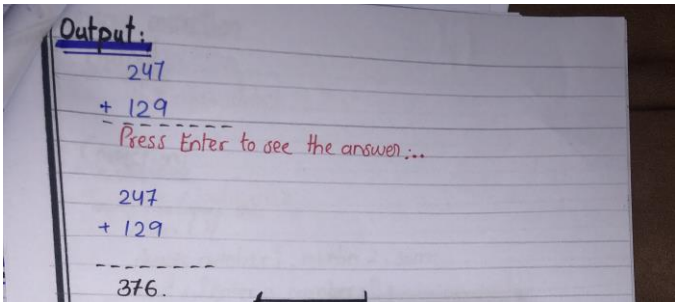
247

+ 129

376

HANDWRITTEN :



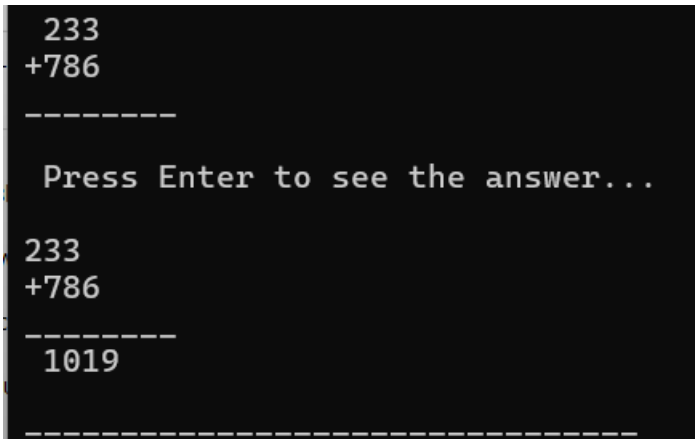


CODE On Desktop :

] Task 6.cpp

```cpp
#include<iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main()
{
    srand(time(0));
    int num1=rand()%900+100;
    int num2=rand()%900+100;
    int answer=num1+num2;
    cout<<" "<<num1<<endl;
    cout<<"+"<<num2<<endl;
    cout<<"________"<<endl;
    cout<<"\n Press Enter to see the answer...";
    cin.get();
    cout<<"\n"<<num1<<endl;
    cout<<"+"<<num2<<endl;
    cout<<"________"<<endl;
    cout<<" "<<answer<<endl;
    return 0;
}
```

OUTPUT :



HOME TASKS:

TASK 01

ERRORS CORRECTION

PROGRAM 01

```
using namespace std;
```

```
void main ()

{

double numberl, number2, sum;

cout "Enter a number: ";

Cin numberl;

cout "Enter another number:”;

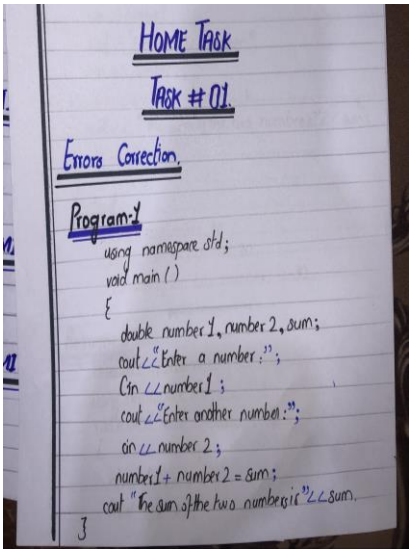
cin number2;

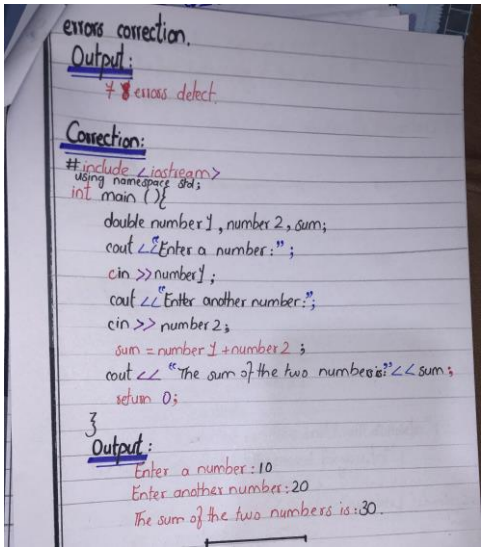
numberl + number2 = sum;

cout "The sum of the two numbers is sum

}
```

HANDWRITTEN :





CODE On Desktop :

Home Task 1.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is "
         << sum << endl;

    return 0;
}
```

OUTPUT :

```
Enter a number: 10
Enter another number: 20
The sum of the two numbers is 30
```

PROGRAM 02

```
#include <iostream>

using namespace std;

void main() {


int numberl, number2;

float quotient;

cout "Enter two numbers and I will divide\n";

cout "the first by the second for you.\n";

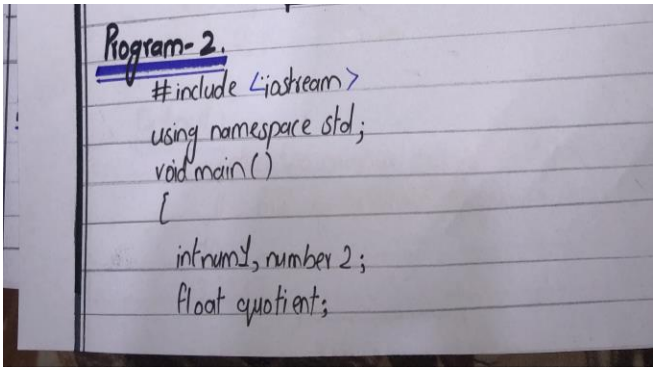
cin >> numberl, number2;

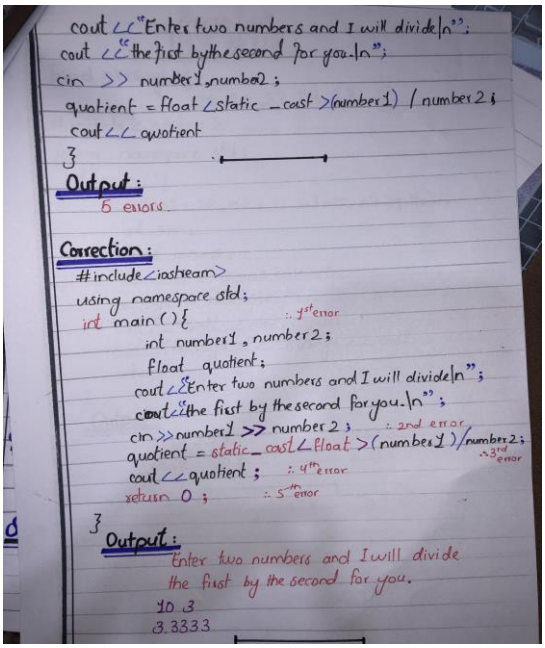
quotient = float<static_cast>(numberl) / number2;

cout quotient

}
```

HANDWRITTEN :





CODE On Desktop :

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << quotient << endl;

    return 0;
}
```

OUTPUT :

```
Enter two numbers and I will divide
the first by the second for you.
10 3
3.33333
```

TASK 02

Average of Values to get the average of a series of values, you add the values up and then

divide the sum by the number of values. Write a program that stores the following values in

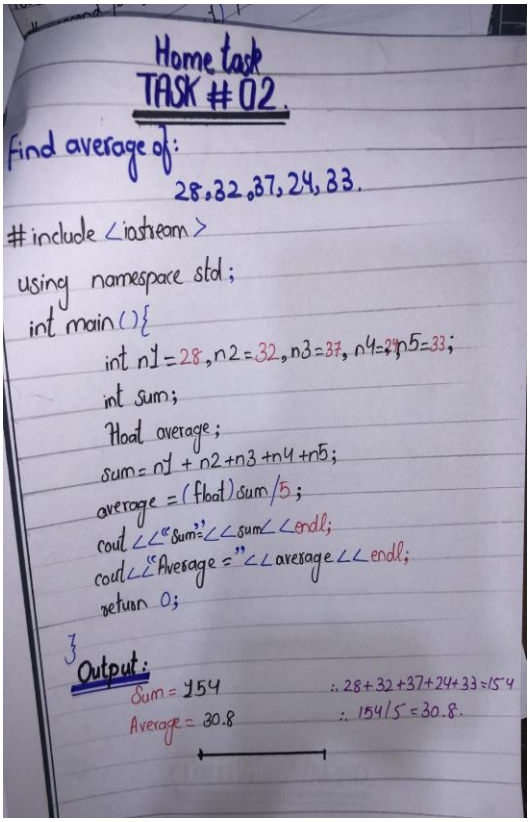
five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of

these five variables and store the result in a separate variable named sum. Then, the

program should divide the sum variable by 5 to get the average. Display the average on the

screen

HANDWRITTEN :



CODE On Desktop :

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n1 = 28, n2 = 32, n3 = 37, n4 = 24, n5 = 33;
    int sum;
    float average;
    sum = n1 + n2 + n3 + n4 + n5;
    average = (float)sum / 5;
    cout<<"Sum="<<sum<<endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

OUTPUT :

```
Sum=154
Average = 30.8

--------------------------------
```